

Formulary – Appendix to Exam

Chapter 1

Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$

Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Present Value of an Annuity: $PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right)$

Future Value of an Annuity:

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Present Value of a Growing Perpetuity: $PV(\text{growing perpetuity}) = \frac{C}{r - g}$

Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)}\right)^N\right)$

Approximate Internal Rate of Return (IRR) via Interpolation (with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Chapter 2

Profitability Index: $\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{NPV}{\text{Resource Consumed}}$

Chapter 3

Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$

Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$

Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$

Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR): $P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$

Chapter 4

Dividend-discount model (1 year): $P_0 = \left(\frac{\text{Div}_1 + P_1}{1 + r_E} \right)$

Total return, capital gain, dividend yield: $r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$

Dividend-discount model (2 years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2 + P_2}{(1 + r_E)^2}$

Dividend-discount model (n years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{\text{Div}_n}{(1 + r_E)^n}$$

DDM, constant growth: $P_0 = \frac{\text{Div}_1}{r_E - g} \quad r_E = \frac{\text{Div}_1}{P_0} + g$

DDM, changing growth rates: $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{\text{Div}_{N+1}}{r_E - g} \right)$

Change in Earnings = New Investment × Return on New Investment

New Investment = Earnings × Retention Rate

Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} \times \text{Return on New Investment}$

$g = \text{Retention Rate} \times \text{Return on New Investment}$

Dividend: $\text{Div}_t = \frac{\text{Earnings}_t}{\underbrace{\text{Shares Outstanding}_t}_{\text{EPS}_t}} \times \text{Dividend Payout Rate}_t$

Chapter 5

Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Variance Estimate Using Realized Returns:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$

$$\begin{aligned} \text{Portfolio beta} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

$$\text{Risk premium on the market} = r_m - r_f$$

$$\text{Risk premium on any asset } i = r_i - r_f$$

$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Chapter 6

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

$$\text{After-tax cost of debt} = r_{\text{debt}} \times (1 - T_c).$$

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$